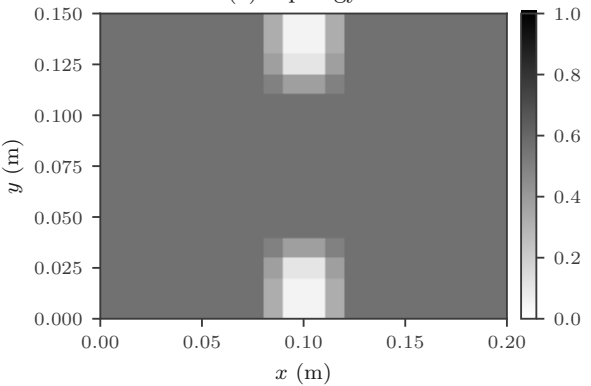
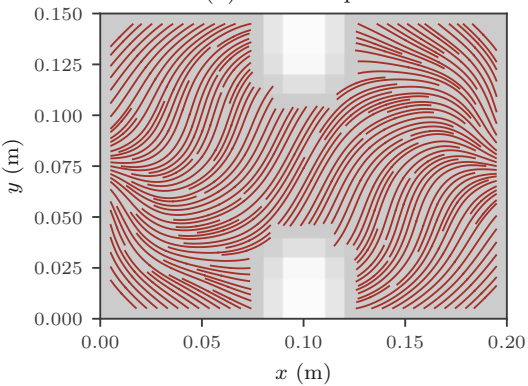
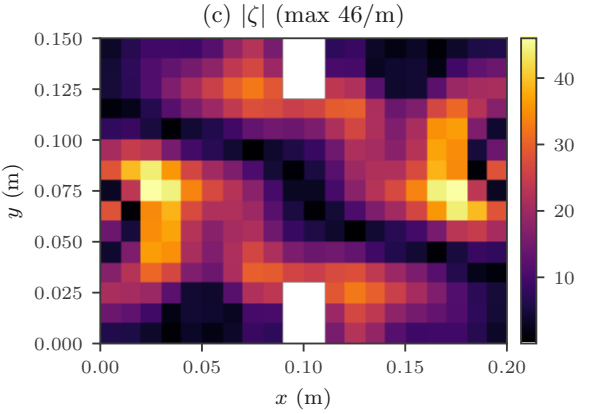
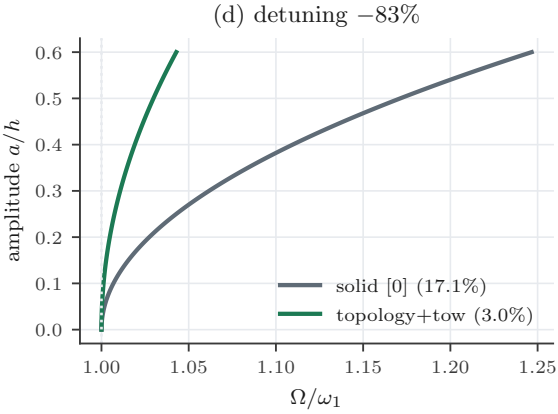

(a) topology z
0.150
0.125
0.100
0.075
0.050
0.025
0.000
y (m)
0.00
0.05
0.10
0.15
0.20
x (m)
1.0
0.8
0.6
0.4
0.2
0.0


(b) fiber toolpaths
0.150
0.125
0.100
0.075
0.050
0.025
0.000
y (m)
0.00
0.05
0.10
0.15
0.20
x (m)


(c) $|\zeta|$ (max 46/m)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
y (m)
0.00
0.05
0.10
0.15
0.20
x (m)
40
30
20
10


(d) detuning −83%
amplitude a/h
0.6
0.5
0.4
0.3
0.2
0.1
0.0
1.00
1.05
1.10
1.15
1.20
1.25
Ω/ω_1
solid [0] (17.1%)
topology+tow (3.0%)